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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9648/02

Paper 2 Structured and Free-Response Questions

15 September 2015

BOOKLET B

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

Please ensure that you also complete BOOKLET A.

For Examiner's Use	
Booklet B [40]	
6	/ 15
7	/ 15
8	/ 10

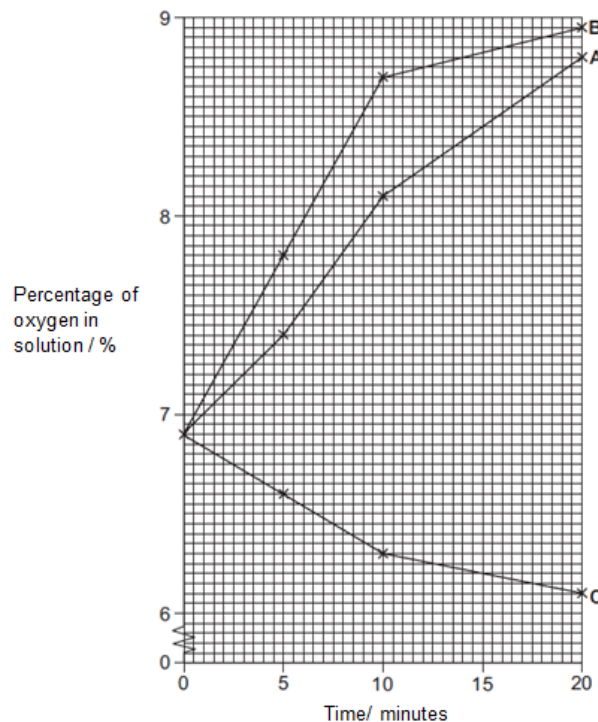
This document consists of **10** printed pages and **0** blank page.

Section B: Structured Questions (80 marks)Answer **all** questions in this section.**Booklet B**For
Examiner's
use**Question 6**

- (a) The following experiment was carried out to investigate the effect of light intensity on the rate of photosynthesis of a water plant, *Elodea*.

- *Elodea* was cut into three pieces, each 10 cm long.
- Each piece of *Elodea* was placed in a glass tube, containing 0.5% sodium hydrogen carbonate solution, which was then sealed with a bung.
- Tube A was placed 10 cm away from a lamp.
- Tube B was placed 5 cm away from a lamp.
- Tube C was placed in a dark room.
- An oxygen sensor was used to measure the percentage of oxygen in the solutions at the start of the experiment and again at 5, 10 and 20 minutes.

The results are shown in **Fig. 6.1**.

**Fig 6.1**

With reference to **Fig. 6.1**,

- (i) Calculate the rate of oxygen production for tube A for the 20 minutes of the experiment. Give your answer to two decimal places. Show your working. [1]

(ii) Compare the results for tubes A and B. [3]

*For
Examiner's
use*

(iii) Describe and explain the results for tube C. [3]

- (b) An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell. **Fig. 6.2** below shows the various experimental systems that are prepared in an attempt to synthesize ATP.

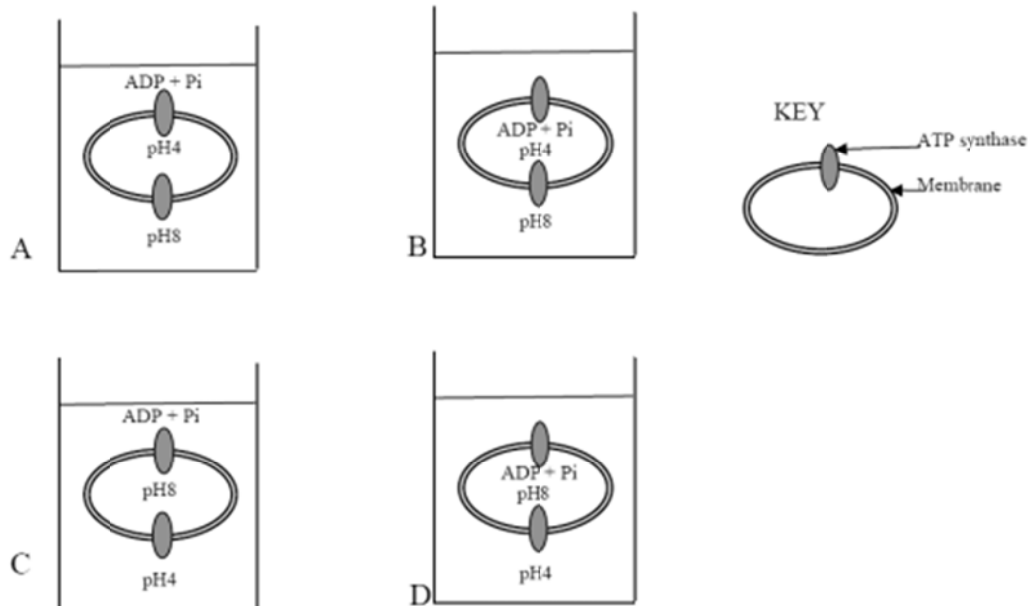


Fig. 6.2

Which experimental system (A to D) shown in **Fig. 6.2** would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer. [3]

(c) Fig. 6.3 outlines the early stages of respiration.

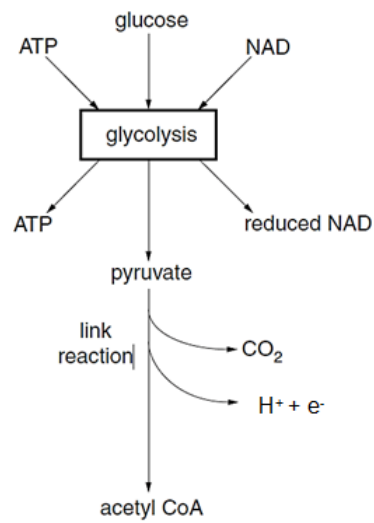


Fig. 6.3

With reference to **Fig. 6.3**,

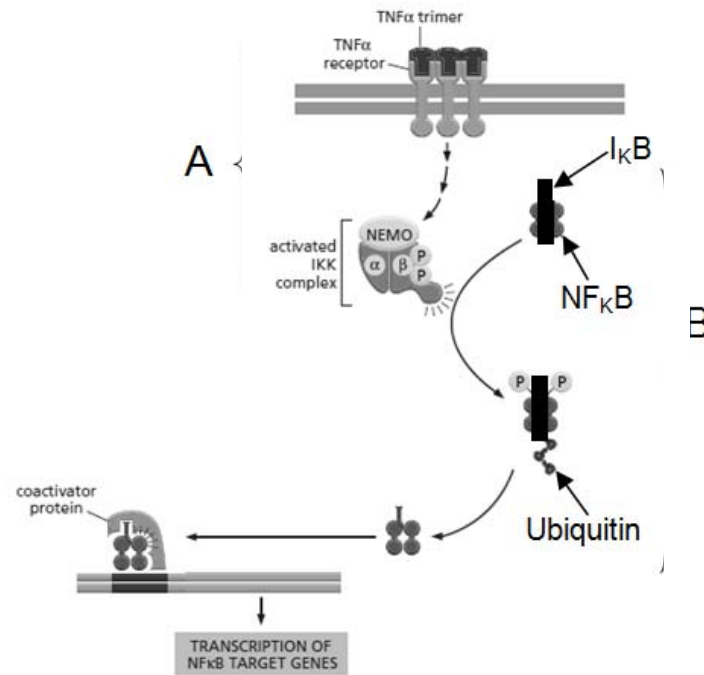
(i) Explain why ATP is needed at the start of glycolysis. [1]

(ii) Explain the fate of the hydrogen atoms released by pyruvate during the link reaction. [4]

Total: [15]

Question 7

- (a) The NF_κB proteins are latent gene regulatory proteins that are present in most animal cells and are central to many stressful, inflammatory, and innate immune responses. NF_κB proteins also have important roles during normal animal development. **Fig 7.1** below shows the cell signaling pathway to activate the transcription of NF_κB gene.

**Fig 7.1**

With reference to **Fig. 7.1**, describe

- (i) Stage A. [2]

- (ii) Stage B. [3]

- (b) Suggest and explain the effect on the gene expression of NF κ B target genes if there was a deletion mutation in the phosphorylation region of protein I κ B. [3]

- (c) Excessive NF κ B signaling is found in a number of human cancers such as breast and liver cancer.

- (i) Explain if NF κ B is a proto-oncogene or tumour suppressor gene. [3]

- (ii) Briefly describe the development of cancer as a multi-step process. [4]

Total: [15]

Question 8

Woolly mammoths are extinct mammals that are related to elephants of the *Mastodon* genus. Woolly mammoths existed around 200,000 years ago during the Ice Age, where the climate changed and temperatures in the Arctic drastically decreased. These mammoths had various adaptations to reduce heat loss, such as having thick fur, small ears and small tails. During the Ice Age, the population of the woolly mammoths was significantly larger than the population of mastodons.

As a result of climate change of increasing temperatures, woolly mammoths went extinct approximately 4000 years ago. However, the population of mastodons increased in number.

(a) Explain why population is considered the smallest unit of evolution. [2]

(b) Explain how natural selection resulted in the population changes of the mastodons. [4]

Scientists studied the haemoglobin of woolly mammoths and Asian elephants to determine whether or not the differences in haemoglobin were part of the mammoth's adaptation to a cold climate.

It is advantageous for Arctic mammals to have haemoglobin whose affinity for oxygen is not significantly affected by changes in temperature. This is achieved by using substances called 'red cell effectors', which can bind to haemoglobin.

Fig. 9.1 shows the effect of temperature on the affinity of oxygen of woolly mammoth and Asian elephant haemoglobin, with and without red cell effectors.

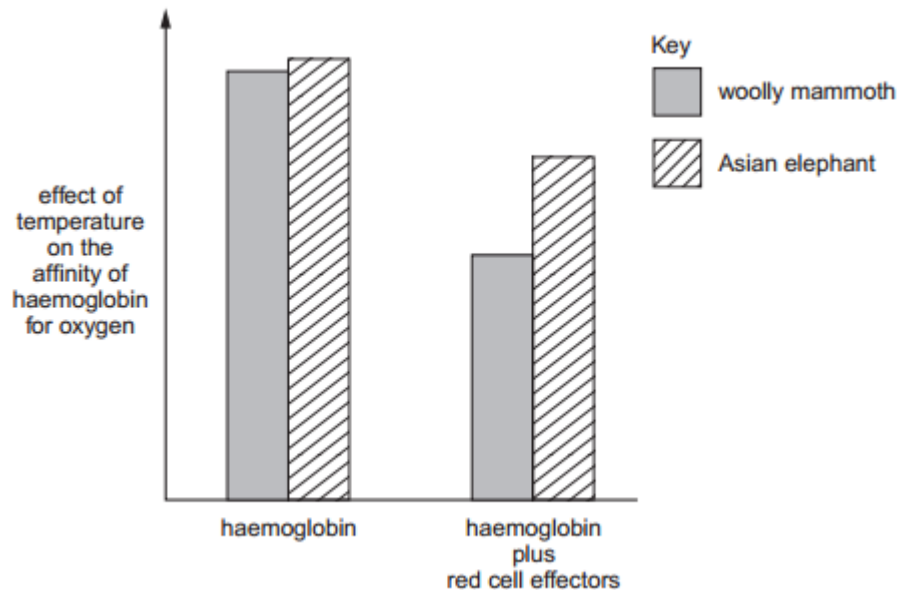


Fig. 8.1

- (c) Suggest how low temperature can affect haemoglobin's activity. [2]

- (d) With reference to **Fig. 8.1**, explain whether or not there is evidence to show that woolly mammoths' haemoglobin is better adapted for a cold climate than Asian elephants. [2]

Total: [10]

Section C: Free-Response Question (20 marks)

Answer only **one** question.

Write your answers on the writing paper provided.

Answer each part (a) and (b) on a fresh piece of writing paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 1

- (a) (i) Compare the structure of glucose and fatty acid molecules. [4]
- (ii) Describe how structure of amylopectin relates to its function. [6]
- (b) Explain with examples, how homology (anatomical, embryological and molecular) supports Darwin's theory of natural selection. [10]

OR

Question 2

- (a) Compare the structure of the tropocollagen and DNA double helix. [10]
- (b) (i) Describe the evolution of mitochondria and chloroplast from prokaryotes using the endosymbiosis theory. [4]
- (ii) Describe and explain the advantages of molecular (nucleotide and amino acid sequences) methods in classifying organisms. [6]

Total: [20]

END OF PAPER



**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H2 BIOLOGY (9648)
PAPER 2 (Booklet B)**

Suggested Answers

Structured Questions Answers

6(a)(i)

$(8.8\% - 6.9\%) / 20 = 0.10\% \text{ per minute}$

6(a)(ii)

Similar start and end point / Similar start but different end point:

From 0-20 mins, percentage O₂ in solution B increased from 6.9% to 8.95% and percentage O₂ in solution A increased from 6.9% to 8.8%.

From 0-20 mins, percentage O₂ in solution B increased from 6.9% to 8.95% which is higher than percentage O₂ in solution A which increased from 6.9% to 8.8%.

1 max

Different rates of increase:

From 0 – 10 minutes, percentage O₂ in solution B increased more steeply from 6.9% to 8.7% / increase of 1.8% but percentage O₂ in solution A increased from 6.9% to 8.1% / increased by 1.2%;

From 10 – 20 minutes, percentage O₂ in solution B increased less steeply from 8.7% to 8.95% / lower increase of 0.25% but percentage O₂ in solution A increased more steeply from 8.1% to 8.8% / higher increase of 0.7%.

6(a)(iii)

From 0 – 20 minutes, percentage of O₂ in solution C decreased from 6.9% to 6.1%;

In the absence of light, no energy was provided for chlorophyll to excite electrons to be passed to the electron transport chain;

There was no photolysis of water and release of oxygen and furthermore, oxygen taken up from the solution to be used up in cellular respiration;

6(b)

Experimental system A; *[must be correct before other points below are awarded]*

Within the membrane, pH 4 has higher concentration of hydrogen ions than outside the membrane of pH 8 outside the membrane which has a lower concentration of hydrogen ions;

This generates a proton motive force / proton gradient / electrochemical gradient. Hydrogen ions can diffuse from the interior of the thylakoid vesicle via the ATP synthase down their concentration gradient out and synthesize ATP using ADP and P_i in the solution outside the vesicle;

6(c)(i)

To phosphorylate the glucose/fructose-6-phosphate / make the glucose molecule more chemically reactive;

6(c)(ii)

H atoms are accepted by NAD^+ to form reduced NAD;

H^+ ions are actively transported into the mitochondrial intermembrane space via proton pump to generate proton gradient;

H^+ ions diffuse via ATP synthase into mitochondrial matrix to generate ATP;

H^+ ions combine with the final electron and hydrogen acceptor oxygen to form water;

7(a)(i)

Ligand $TNF\alpha$ trimer has a complementary structure to the $TNF\alpha$ receptor and binds to it, activating it;

Activation of the $TNF\alpha$ receptor results several steps and finally the formation of an activated IKK complex;

7(a)(ii)

Activated IKK complex phosphorylates $I\kappa B$ and results in ubiquitin tagging $I\kappa B$;

$I\kappa B$ acts as a repressor / inhibitor gets degraded by proteasome;

$NF\kappa B$ dissociates and becomes activated, binds to DNA / acts as a transcription factor;

7(b)

There would be no gene expression of $NF\kappa B$ target genes;

$I\kappa B$ cannot be phosphorylated and ubiquitin cannot be tagged onto the protein to mark it for degradation;

$NF\kappa B$ cannot dissociate and remains inactivated, and cannot bind to DNA;

7(c)(i)

Proto-oncogene which normally regulates cell growth and differentiation;

Proto-oncogene becomes an oncogene due to overexpression resulting in excessive cell growth and division which causes cancer;

Tumour suppressor gene under expression would result in cancer / Tumour suppressor gene over expression would not result in cancer;

7(c)(ii)

Single cell becomes genetically mutated, and begins to divide uncontrollably;

One of the descendent cells mutates further and this mutation is passed to daughter cells;

Cancer cells are derived from a single cell that has undergone a series of 5-8 mutations;

Other mutations may result in the cell being able to undergo angiogenesis (develop new blood vessels) / metastasis (able to invade other tissues);

8(a)

Individuals accumulate mutations and change phenotypes because of the environment but do not evolve during their lifetimes;

Evolution requires descent with modification, it can only be observed in the next generation at a population level;

8(b)

Change in climate / warmer climate acted as selection pressure;

Mastodons with thin fur and large ears were at a selective advantage as they were better adapted for warm climate as compared to the thick fur woolly mammoths;

Mastodons had a higher chance of survival and thus reproduce to give viable offspring, passing the advantageous alleles to them;

Over time, allele frequencies for fur and ears changed such that mastodons population increased in number while woolly mammoths population decreased in number;

8(c)

At low temperature, haemoglobin is inactivated as it is a protein;

It has low kinetic energy and the frequency of effective collision between haemoglobin's binding site and oxygen is low hence the activity of protein is low; R! Active site

8(d)

Yes there is evidence; *[must be correct before other points below are awarded]*

The presence of red cell effectors reduces the effect of temperature on the affinity of haemoglobin for oxygen to a larger extent in the woolly mammoth than Asian elephant;

Essay Answers

1(a)(i) Compare the structure of glucose and fatty acid molecules. [4]

	glucose	Fatty acid
1	Can either be a linear structure or ring which is more stable	Linear structure
2	^1C contains an aldehyde group, while $^2\text{C} - ^5\text{C}$ each possess a hydroxyl group (-OH).	Consist of a <u>carboxyl</u> group and R group which is a long hydrocarbon chain of varied length.
3	Smaller molecule.	Larger molecule.
4	Isomers: α and β glucose and fructose.	No isomer
	Max 4	
S	Both contain the elements C, H and O	

1(a)(ii) Describe how structure of amylopectin relates to its function. [6]

1. starch is able to store large amounts of energy as it is made of several hundred / thousands of glucose monomers linked together by glycosidic bonds;
2. amylose has α (1 $\rightarrow$ 4) glycosidic bonds and amylopectin has both α (1 $\rightarrow$ 4) and α (1 $\rightarrow$ 6) glycosidic bonds;
3. α (1 $\rightarrow$ 4) glycosidic bonds gives rise to unbranched chains of glucose monomers which then coil helically due to intrachain hydrogen bonding $\rightarrow$ compact for storage;
4. ref to α (1 $\rightarrow$ 6) glycosidic bonds giving rise to branched structure $\rightarrow$ compact for storage;
5. α (1 $\rightarrow$ 4) glycosidic bonds can be easily broken down by enzymes present in plants and animals releasing glucose for respiration;
6. branched structure allows for quick break down / hydrolysis of starch due to action of debranching enzymes on α (1 $\rightarrow$ 6) glycosidic bonds;
7. Starch is a large molecule with OH groups occupied in intrachain H-bonding $\rightarrow$ insoluble; therefore starch is osmotically inactive / will not affect osmotic concentration in cells $\rightarrow$ can be stored in large amounts in cells;

6 max.

1(b)	Explain with examples, how homology (anatomical, embryological and molecular) supports Darwin's theory of natural selection.
Marking Point	Answer
1	Homology is defined as the similarities between individuals of the same species or among different species that arose from common ancestry;
2	In anatomical homology, comparison of body structures between species provides evidence of inheritance from a common ancestor, and that modifications are adaptations to the special needs of the organism;
3	For example, the forelimbs of all mammals, including humans, cats, whales and bats show the same arrangement of bones from the shoulder to the tips of the digits, even though these appendages can have very different functions;
4	Homologous structures such as the different forelimbs support Darwin's theory of evolution by natural selection whereby new species evolve by descent with modification from their common ancestor;

5	Analogous structures such as the membrane of skin on either side of body are found in both the Australian Sugar Glider and North American Flying Squirrel. This also supports Darwin's theory of evolution by natural selection as the environment selects for the genes which are the most advantageous hence the phenotypes converge;
6	In embryological homology, comparison of early stages of animal development reveals additional anatomical homologies not visible in adult organisms. Embryos that look very similar during the earlier stages of development have a more recent common ancestor;
7	For example, all vertebrate embryos have a tail posterior to the anus, as well as gill pouches. These embryonic structures develop into homologous structures with very different functions, such as gill slits in fishes and parts of ears and throat in humans;
8	Molecular homology is the similarities among organisms at the molecular level. All forms of life use the same genetic machinery of DNA and RNA, and the genetic code is essentially universal;
9	Two organisms with high molecular homology in their genes essential for development and also important functions reflect that they share a common ancestor;
10	For example, haemoglobin amino acid sequence, humans and rhesus monkey could be closely related as there is only a difference of 8 amino acids in the haemoglobin polypeptide sequence;
11	AVP, examples that are sound but not mentioned above;

10 max

2(a) Compare between structure of the tropocollagen and DNA double helix. [10]

	tropocollagen	DNA double helix
1	Made up of 3 polypeptide chains wind around each other to give a triple helix	Made up of 2 DNA strands
2	Each polypeptide chain contains about 1000 amino acids	Each strand contains a lot more deoxynucleotide.
3	The amino acid is mostly proline and glycine, and thus very repetitive	Coding DNA is made a unique and non-repetitive sequence of deoxynucleotide.
4.	Glycine, being small in size, is the key to allows the three strands to lie close together to form a tight coil.	Base pairing between a purine and a pyrimidine is essential to keep the width of the DNA double helix at a constant 2nm.
5	Each completed tropocollagen lies parallel in a regularly staggered arrangement with other triple helices to form microfibrils followed by fibrils, joined by covalent bonds between neighbouring triple helices	DNA double helix does not join with other double helix to form a thicker and more stable structure.
6	Fibrils unite to form fibres resulting in great tensile strength.	DNA double helix wind around histone into nucleosome, which go through further condensation to maintain its integrity during nuclear division.
	Both are made up more than one molecule;	
	Both structures are made up of chains and strands held together by hydrogen bonds;	
	Amino acid sequence in tropocollagen is repetitive. And deoxyribonucleotide sequence in non-coding DNA is repetitive too.	

2(b)(i)	Describe the evolution of mitochondria and chloroplast from prokaryotes using the endosymbiosis theory. [4]
Marking Point	Answer
1	Endosymbiosis theory hypothesises that mitochondria and chloroplasts were formerly small prokaryotes that began living within larger cells;
2	The prokaryotic ancestors of mitochondria and chloroplasts had genes which allowed them to be aerobic and photosynthetic gained entry to the host cell as undigested prey or internal parasites;
3	The host cell was probably a heterotroph, an organism that eats other organisms and could use nutrients released from the prokaryotic ancestors;
4	The selection pressure of increasing oxygen and sunlight, selected for those host cells that could effectively utilise oxygen and light to produce more energy;
5	Over a long period of time, the prokaryotic ancestors entered into a mutually beneficial relationship with the host cell and later became the modern eukaryotic cells today;
6	The early origin of mitochondria is supported by the fact that all eukaryotes studied so far have either mitochondria or modified versions of them;

2(b)(ii)	Describe and explain the advantages of molecular (nucleotide and amino acid sequences) methods in classifying organisms. [6]
Marking Point	Answer
1	Molecular evidences are quantifiable as compared to morphological evidence, which are qualitative because they involve visual observation of a phenotype;
2	For example, percentage homology in DNA and amino acid sequences between species as compared to shape of head;
3	Thus molecular evidences are unambiguous and objective while morphological evidences are subjective;
4	Molecular evidences are open to statistical analysis while it is very difficult to do so for morphological evidence as they are qualitative evidences;
5	Statistical analysis can include chi-square or t-test to determine goodness of fit or significant difference between samples;
6	Morphological evidence may be erroneous because morphological similarities between two species may be due to convergence evolution when the species are subjected to similar selective pressure but in fact they are from two different ancestral lineages;
7	For example, sharks and dolphins both have streamlined bodies due to need to swim faster in the sea however, dolphins are mammals while sharks are fishes;